Course Syllabus

General Information

Faculty	Information Technology		
Department	Software Engineering		
Semester	Second	**Academic Year**	2025/2026

Course Details

Course Title	Software Architecture and Design			**Course No.**	**A0423501**	
Type of Learning	■ On- campus ☐ Blended (……..….. Platform) ☐ Online (……..…. Platform)					
Credit Hours	3	**Theoretical**	3	**Practical**	0	
Course Level (according to JNQF)			7	**JNQF Hours***	12.15	
Pre-requisite	**A0422502 – Software Requirements Engineering**			**Co-requisite**		

Course Instructor Information

Instructor Name	Dr. Shadi Ettantawi		
Instructor E-mail	s.ettantawi@meu.edu.jo		
Course Time	Sunday – Tuesday – Thursday: 12:00 – 13:00		
Office Hours			
Office Number	N143	**Office Phone**	
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook
Fundamentals of Software Architecture, Mark Richards & Neal Ford, O'Reilly Media, Inc, 2020.
2. References
Clean Architecture A CRAFTSMAN'S GUIDE TO SOFTWARE STRUCTURE AND DESIGN, Robert C. Martin, Pearson Education, 2018.

Teaching Methods

1. Lectures	**2.** Case Studies	**3.** Interactive demonstrations
4. Assignments	**5.** Discussion & Questions	**6.** Project

Course Description

This course offers a comprehensive understanding of software architecture and design principles, techniques, and best practices for building maintainable, robust, and scalable systems.

Students will explore architectural styles while engaging in lectures, assignments, and projects to develop problem solving skills, scenario analysis, solution design, and effective team work.

Throughout the course, students will analyze real-world software systems, understand design decisions and trade-offs, and ultimately become equipped to tackle diverse software design challenges and create innovative and maintainable solutions.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	**Describe** various software architectures concepts and design alternatives.	PLO1	I		
2	**Explain** different design principles and trade-offs encountered in software architecture and design.	PLO1	I		
3	**Assess** modern software architecture styles with respect to real-world systems.	PLO2		P	
4	**Design** and present a suitable software architecture for a given system within specific requirements.	PLO2		D	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture Type	Lecture No.		Assessment Tool	Summative/ Formative
1	**Introduction:** - Defining Software Architecture -Laws of Software Architecture	Lecture (1)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
2	-Architecture Versus Design -C4 model	Lecture (1)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	1	**Quiz**	**Summative**
3	**Modularity:** - Cohesion - Coupling - Modules and Components	Lecture (1)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	1	**Discussion & Questions**	**Formative**
4	**Architecture Characteristics:** - Fundamentals - Categories - Trade-offs	Lecture (1)	Lecture On- campus	2	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	2	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	2	**Discussion & Questions**	**Formative**
5	**Architecture Styles Overview:** - From Big Ball of Mud, to Modern Distributed Systems.	Lecture (1)	Lecture On- campus	2	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	2	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	2	**Quiz**	**Formative**
6	**Layered Architecture:** - Topology - pros & cons - 2 tier and 3 tier implementations	Lecture (1)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	4	**Discussion Questions &**	**Summative**
7	**Pipeline Architecture:** - Topology - pros & cons - Implementation	Lecture (1)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	4	**Discussion & Questions**	**Formative**
8	**Mid Exam**			1, 2, 3		**Summative**
9	**Microkernel Architecture:** - Topology - pros & cons - Implementation	Lecture (1)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	4	**Discussion & Questions**	**Formative**

10	**Service-Based Architecture:** - Topology - pros & cons	Lecture (1)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	4	**Assignment**	**Summative**
11	**Event-Driven Architecture:** - Topology - pros & cons	Lecture (1)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	3	**Quiz**	**Summative**
12	**Peer-to-Peer architecture:** - Topology - pros & cons - Implementation	Lecture (1)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	3	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	4	**Discussion & Questions**	**Formative**
13	**Presentations**	Lecture (1)	Lecture On- campus	4	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	4	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	4	**Project**	**Summative**
14	**Diagramming and Presenting Architecture**	Lecture (1)	Lecture On- campus	4	**Discussion & Questions**	**Formative**
		Lecture (2)	Lecture On- campus	4	**Discussion & Questions**	**Formative**
		Lecture (3)	Lecture On- campus	4	**Discussion & Questions**	**Summative**
15	**Diagramming and Presenting Architecture**	Lecture (1)	Lecture On- campus	4	**Discussion**	**Summative**
		Lecture (2)	Lecture On- campus	4	**& Questions**	**Summative**
		Lecture (3)	Lecture On- campus	4	**Discussion & Questions**	**Formative**
16	**Final Exam**		Lecture On- campus	1, 2, 3, 4		**Summative**

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Mid Exam	25%	Week 8
Quizzes	10%	Week 5, 11
Assignment	10%	Week 10
Project	15%	Week 13
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	▪ No makeup quizzes. ▪ 2 quizzes are given. Each Quiz is out of 5.
Exams	▪ The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	▪ Makeup exam should not be given unless there is a valid excuse.
Drop Date	▪ The last day to drop the course is according to the University regulations.
Academic Honesty	▪ Cheating or copying on exams or quizzes is an illegal and unethical activity. ▪ Standard MEU policy will be applied. ▪ All graded assignments must be your own work (your own words).
Attendance	▪ Excellent attendance is expected. ▪ MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. ▪ If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	▪ It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	▪ The student should expect to spend 6 hours per week as an average workload.
Graded Exams	▪ Instructor should return exam papers graded to students within one week after the exam date.
Participation	▪ Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. ▪ Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	▪ Assignments are as mentioned in the previous table. ▪ Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. ▪ More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

* **JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3	13.5	40.5
Tutorials	4	5	20
laboratory	0	0	0
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)			
Final Exam	2	1	2
Quizzes	1/2	2	1
Self-Learning Activities			
Assignments/ Homework	3	5	15
Case Studies	3	1	3
Presentations	3	1	3
E-learning			
Extra Readings	3	12	36
Total Hours	**121.5**		
JNQF Hours	**Total Hours/10 = 12.15**		